

SQL fundamentals

Exercise 2. SQL CASE statements

1. Products - Price Category

SQL

```
SELECT product_name,  
       price,  
       CASE  
         WHEN price > 1000 THEN 'Expensive'  
         WHEN price BETWEEN 100 AND 1000 THEN  
           'Mid-range'  
         ELSE 'Budget'  
       END AS price_category  
FROM products;
```

Results

Product_name	price	Price-category
Laptop	1200.00	Expensive
Phone	500.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

2. Orders - Order Value Category

SQL query

```
SELECT customer_name,  
       amount
```

```
CASE
```

```
WHEN amount >= 1000 THEN 'High Value'
```

```
WHEN amount BETWEEN 500 AND 999.99 THEN
```

Medium Value

```
ELSE 'Low Value'
```

```
END AS order_value_category  
FROM orders;
```

Results

customer_name	amount	order_value_category
Alice	150.00	Low Value
Bob	560.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

3. employees - Position Level

SQL query

```
SELECT employee-name,  
       department,  
       salary
```

CASE

WHEN department = 'IT' AND salary > 8000 THEN

'Senior IT'

WHEN department = 'HR' AND salary > 5500 THEN

'Experienced HR'

ELSE 'Staff'

END AS position_level.

FROM employees;

Results

employee-name	department	salary	position-level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4. Students - Grades

SQL

```
SELECT student_name,
       score
```

CASE

WHEN score >= 90 THEN 'A'

WHEN score >= 80 THEN 'B'

WHEN score >= 70 THEN 'C'

WHEN score >= 60 THEN 'D'

END CASE AS grade

FROM student;

Student_name	score	grade
Anne	92	A
Ben	76	C
Caru	59	F
David	83	B
Ella	68	D

5. Deliveries - Performance

```
SELECT delivery_id,
```

delivery_time_minutes

CASE

WHEN delivery_time_minutes <= 30 THEN 'fast'

WHEN delivery_time_minutes <= 60 THEN 'on time'

ELSE 'late'

END AS performance

FROM deliveries

Results

Delivery-id	delivery-time-minutes	Performance
1	45	On Time
2	80	Late
3	30	fast
4	65	Late
5	100	Late

6. Tickets - Priority Labels

SQL query

SELECT issue_type
priority

CASE

WHEN priority = 3 THEN 'High'

WHEN priority = 2 THEN 'Medium'

WHEN priority = 1 THEN 'Low'

END AS priority_label

FROM tickets

Results

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	medium
Email error	2	medium
Password reset	1	Low

7. Attendance - status

```

SELECT student_id
      (days_present * 100.0 / total_days) AS
attendance_percentage
CASE
WHEN (days_present * 100.0 / total_days) >= 90
THEN 'Excellent'
      WHEN (days_present * 100.0 / total_days) >= 70
THEN 'Good'
      ELSE 'Needs Improvement'
END AS attendance_status
FROM attendance;
  
```

Results

Student id	attendance_percentage	attendance_status
1	90%	Excellent
2	60%	Needs Improvement
3	96%	Excellent
4	50%	Needs Improvement
5	100%	Excellent

8. Product Inventory - stock status

```

SELECT product_id,
      stock_qty
CASE
WHEN stock_qty = 0 THEN 'out of stock'
      WHEN stock_qty BETWEEN 1 AND 5 THEN
'Low Stock'
  
```

~~ELSE~~ ELSE 'in stock'
 END IF stock_status
 FROM product_inventory ;

Results

Product-Id	stock_qty	stock_status
1	5	Low stock
2	0	Out of stock
3	25	in stock
4	10	in stock
5	3	Low Stock

9. Classes - class size

SELECT subject,
 enrolled_students,

CASE

WHEN enrolled_students >= 25 THEN 'Large'

WHEN enrolled_students BETWEEN 10 AND 24

THEN 'medium'

ELSE 'small'

END AS class_size_category

from classes ;

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	medium
Art	5	small
History	20	medium

Q. Payment - Discount eligibility

SELECT payment_id
payment_method,
amount

FROM

WHERE payment_method = 'cash'

AND amount >= 200

THEN 'Eligible for Discount'

ELSE 'not eligible'

END AS discount_eligibility
FROM payments

Results

payment_id	payment_method	amount	discount_eligible
1	Card	50	not eligible
2	Card	200	not eligible
3	Card	150	Not eligible
4	Paypal	75	not eligible
5	Cash	300	not eligible